ThaiSLM Reverse — Result Report v2 (text / speech → Thai Sign Language avatar)

Scope (reverse_text2sign/prompt/prompt_2.md): improve v1 (`result_v1.md`) on the seven gaps it reported and under the six v2 constraints — sign correctness, language–motion separation, data provenance, motion continuity, uncertainty and reproducibility — measured against v1 **on the same tests**, with held-out data kept apart from development data. Reverse part only (`reverse_text2sign/`), production stack on **:8081**, cloud cost ≤ $5–6.

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0. TL;DR

question	answer
What changed for the user?	A **Grammar Format** switch (TSL51 rules ↔ Normal Thai order, same signs), **live text while speaking**, a **reliability check** that refuses to sign an invented transcript, **correction suggestions** the person accepts, a **provenance box** (where every sign comes from, its quality, licence and any repair), all **5 characters** working, the wai palms meeting instead of crossing, hands that reach the face.
Were there hidden v1 defects?	**Yes, four, found by the v2 tests and fixed:** (1) in the browser the thumb rotations landed on the wrong joints (three-vrm renames VRM 0.x thumbs) — Python metrics never saw it; (2) face-contact signs were ~**14–20 cm in front of the face** (correct only from the front camera); (3) the only recorded **ตา** is *grandfather*, so "my **eye** hurts" signed GRANDFATHER; (4) VRM 1.0 models with rest rotations (gojo, yuta) crashed the rig loader.
Language	TSL51 regression **62/62** (unchanged); 148 of 149 everyday sentences identical to v1 (the 149th now correct after a fix). **Frozen held-out set** (85 new sentences, written before any change): sentences passing all checks **82.4 % → 87.1 %** (homographs 64 → 91 %, English 92 → 100 %).
Motion	Palm-to-palm contact: palms facing within **8°** (v1: crossed — the tracker merged the two hands); hand-through-hand frames 41 → 27 over 96 sentences; face-located signs now reach the face (ห้องน้ำ: 18.7 cm → ~3 cm in front of the nose). Strokes are provably untouched by the transitions (test).
Face / head	Every sentence-level mark is **scoped and cites data**: WH brows from the 2nd sign (TSL51), yes/no head tilt on the last sign (4 recorded phrases, p = 0.03), negation head-shake on the negation sign (109 dictionary clips, 12 signers, p = $6{\cdot}10^{-17}$); **topic marking not added** — the data show none (p = 0.53).
Speech	GPU: Whisper-th **medium** (real speech CER **3.8 %** vs 6.1 % small; 0.7 s per short utterance on the RTX 2050). CPU: **small** (medium is 6.9 s — too slow). VAD pause-chunking was **tested and rejected for accuracy** (6.1 → 6.6 % CER on real speech) and is used only for live partial text. Automatic correction **hurt** real speech (7 fixes vs 24 harms) → shipped as suggestions only.
Coverage	No clean adult recording of ทำ · เสื้อ · ราคา · ลดราคา · จริง · ทุก · โมง exists in any source I can reach (800+ YouTube results searched; the only เสื้อ lesson is a school student — rejected under the v1 rule). These remain fingerspelled; a 10-word recording list is in §10.
Cost	**$0 cloud** (no Vertex job, no GCS object — verified empty after the work). GPU work ran on the local RTX 2050.

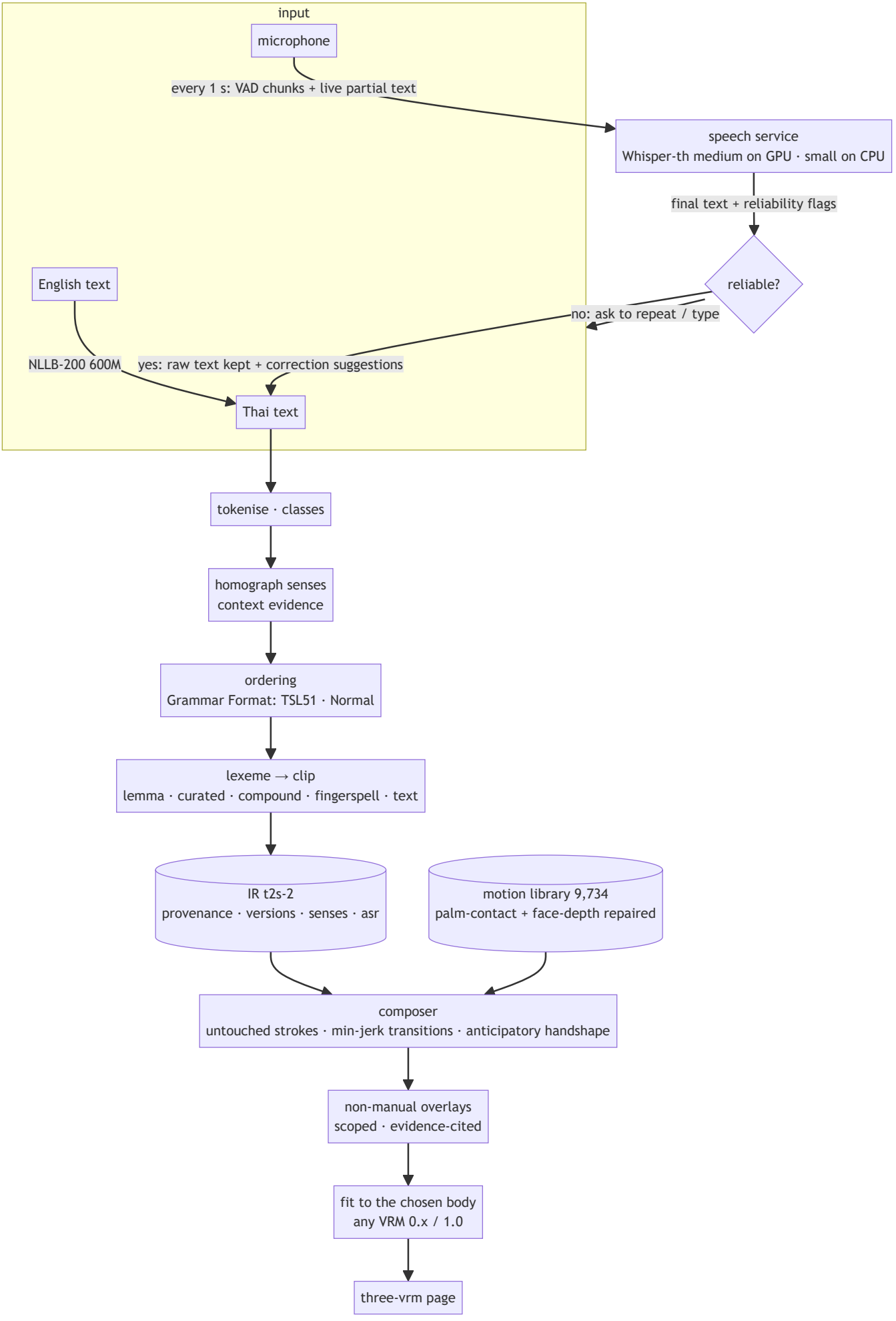
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1. The v2 constraints → what the system does

#	constraint (prompt v2)	design answer	where
1	sign correctness	four motion origins kept apart in the IR: `recorded_clip`, `recorded_phrase`, `composed` (a compound / fixed expression assembled from recorded signs), `fingerspelling`, `text`	`t2s/ir.py`, `t2s/text/senses.py`

#	constraint (prompt v2)	design answer	where
		(nothing signed). Homographs never borrow the other meaning's sign.	
2	language–motion separation	the language stage only asks "is there a recording for this concept"; the motion stage never changes the gloss list. Grammar modes change order only; motion fixes change no translation (148/149 everyday sentences identical to v1).	`t2s/text/` · `t2s/motion/`
3	data provenance	per sign: dataset, clip, signer, tier, landmark-quality metrics from the audit, retargeting error, repairs applied, licence, source link — in the IR and the page's **provenance box** (bottom left)	`t2s/text/lexemes.py`, UI
4	motion continuity	strokes are copied untouched (test); transitions are the only synthesised frames; contact constraints only where the evidence says the hands touch	`t2s/motion/compose.py`, `clean.py`, `retarget.py`
5	uncertainty	ambiguous word → recorded meaning + warning + lower confidence; meaning without a sign → fingerspelling with the reason; unreliable transcript → not signed; every non-manual mark lists its evidence	IR `sense`, `warnings`, `nmm`
6	reproducibility	v1 kept as a git worktree and run on every new test; held-out language and speech sets frozen before changes; IR carries content hashes of lexicon, motion library, timing, rig and rules	`ir.versions`, §4–§8

2. Blueprint (v2)


input
microphone
every 1 s: VAD chunks + live partial text
speech service
Whisper-th medium on GPU · small on CPU
final text + reliability flags
reliable?
no: ask to repeat / type
English text
NLLB-200 600M
yes: raw text kept + correction suggestions
Thai text
tokenise · classes
homograph senses
context evidence
ordering
Grammar Format: TSL51 · Normal
lexeme → clip
lemma · curated · compound · fingerspell · text
IR t2s-2
provenance · versions · senses · asr
motion library 9,734
palm-contact + face-depth repaired
composer
untouched strokes · min-jerk transitions · anticipatory handshape
non-manual overlays
scoped · evidence-cited
fit to the chosen body
any VRM 0.x / 1.0
three-vrm page

Why still concatenative, no new model. The prompt asks to add AI only with a clear role. The two places where a model could help were tested and the evidence decided: a larger ASR model where a GPU exists (it is better on real speech), and **no** LLM for TSL ordering (there is no parallel data beyond TSL51 to check its output against; the Grammar Format switch lets experts compare two transparent orderings instead). GPU training was never needed.

KPIs (defined from the prompt before the work) and where they ended

KPI	target	v1	v2
regression tests	100 % of supported cases	23 tests, 28/29 run today (gojo crash)	**44 / 44**
TSL51 gold order (dev)	no regression	62/62	**62/62**
everyday sentences (149)	no regression	—	**148 identical, 1 corrected**
held-out sentences, all checks pass	better than v1	82.4 %	**87.1 %**
invented / wrong-meaning signs	0	ตา (eye) → GRANDFATHER	**0 found** (test)
motion validity (NaN, non-unit quaternions, bone length)	0	0	**0** (test)
hand-through-hand frames (96 sentences)	lower	41	**27**
palm-to-palm contact (ขอบคุณ)	palms facing	crossed	**8°**
face contact (ห้องน้ำ nose)	reaches the face	18.7 cm in front	**≈ 3 cm**
Thai speech CER, real speech (FLEURS)	lower	6.1 % (small)	**3.8 % (GPU) / 6.1 % (CPU)**
text → motion latency	no regression	59 ms p50	**53 ms p50**
reproducibility	versions recorded	—	**IR content hashes**

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3. Language — TSL structure

3.1 What was wrong and what changed

change	evidence / why	effect
Homograph senses (`senses.py`): ตา (grandfather ↔ eye), มัน (yam ↔ it), กา (crow ↔ kettle), เขา (pronoun ↔ horn). A meaning without a recording is fingerspelled, never replaced by the other meaning's sign; no context cue → recorded meaning with a warning and confidence 0.6	TTRS / th-sl definitions checked entry by entry: the only ตา clip is *พ่อของแม่ / Grandpa*	"ตาของฉันเจ็บ" no longer signs GRANDFATHER
expletive vs pronoun มัน	v1 dropped มัน before any verb ("มันชอบนอน" lost its subject)	dropped only before a state ("มันร้อน")
hair vs I (ผม)	"ผมของเธอยาว", "ตัดผม" read as /	ผม before ของ / after ตัด·สระ·หวี… = hair
name boundary after ชื่อ	"หมอชื่อวิชัยใจดีมาก" spelled the whole tail	the name stops at particles / degree words / known adjectives
lexeme layer	concept, sense, variant and clip were one string	`lexeme {concept_id, sense, variant_id, n_variants, variants}`; experts can switch to another recorded variant in the provenance box

3.2 Grammar Format (user-selectable)

`TSL51` = v1's rules (time first · object/topic before subject · head-final verb chain · WH last · negation after the predicate · possessor first). `ปกติ (ไทย)` = **the same signs** — same dropped words, same lexical choices, same sentence type and face — in Thai word order, and without the recorded-phrase shortcut (a recorded phrase is in TSL order). Only the order differs, so the two can be compared on any sentence; the IR records the mode and the rule version (test: `test_grammar_modes_same_signs`).

Example: *ฉันไม่ชอบกินผัก* → TSL51 **ผัก · ฉัน · กิน · ไม่ชอบ** · Normal **ฉัน · ไม่ชอบ · กิน · ผัก**.

3.3 Held-out evaluation (frozen before the changes, `tests/language_heldout.tsv`)

85 sentences (73 Thai, 12 English) never used to develop a rule, by category, each with checks that need no gold gloss sequence: sentence type, a sign present / absent, position of a sign (TSL51 mode), fingerspelled names. Order checks test that the declared rules are applied *consistently* to new sentences — not that the rules are right (only Deaf experts can judge that). Same checker for v1 (git worktree) and v2.

category (n)	v1 all-pass	v2 all-pass (TSL51)	v2 all-pass (Normal)
WH questions (10)	100 %	100 %	100 %
yes/no questions (8)	87.5 %	87.5 %	87.5 %
negation (9)	88.9 %	88.9 %	88.9 %
time (8)	87.5 %	87.5 %	87.5 %
numbers (7)	85.7 %	85.7 %	85.7 %
names (6)	66.7 %	66.7 %	66.7 %
homographs (11)	**63.6 %**	**90.9 %**	90.9 %
long sentences (8)	62.5 %	62.5 %	62.5 %
dropped subject / particles (6)	83.3 %	83.3 %	83.3 %
English (12)	**91.7 %**	**100 %**	100 %
all (85)	**82.4 %** (161/178 checks)	**87.1 %** (165/178)	87.1 % (145/158)
words signed by recordings	96.4 %	95.8 %	95.8 %

Honest notes: (a) coverage is slightly *lower* because "ตา (eye)" is now fingerspelled instead of signed with the wrong sign; (b) one held-out sentence (h53 "มันชอบนอน") uses the pattern I fixed while testing my own example *มันชอบนอน* — the same sentence — so 1 of the 4 newly passing items is contaminated; (c) **8 of the 11 remaining failures are the checks being too literal**, not system errors: the dictionaries hold the compound as one sign (อ่านหนังสือ, ไม่เข้าใจ, อยู่บ้าน, สามสิบ, ฝนตก, ลูกชาย, ทำอาหาร, ง่วง); (d) the real failures were a name glued to the next adjective (h48) and ผม read as *I* in "ผมของเธอ" (h50) — both fixed **after** the frozen run (so not counted above) — and a cat named ส้ม signed as ORANGE (h47, open: a one-word name that is also a dictionary word).

3.4 Regression

TSL51 62/62 (development set, as in v1 — not a generalisation test). 149 everyday sentences: 148 identical gloss sequences to v1; the 149th (*This is delicious* → MT "มันอร่อยมาก") regressed during development (a context cue revived the dropped expletive as *yam*) and was fixed; coverage TH 97.6 % / EN 97.8 % of words signed, unchanged.

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4. Lexicon, selection and provenance

- **Concept · sense · variant · clip separated** (`lexemes.py`): 9,734 library entries → concept groups; school / regional variants (Setsatian, Ratchasuda, "form n") are variants of one concept, parentheses that are not variant tags are senses.
- **Selection order**: meaning (sense, from context) → the concept's default variant (v1's consensus medoid) → its clip; other variants are listed, never substituted silently; **no nearest-meaning fallback** (v1 measured: 9 % recall at 90 % precision — it would mostly show a wrong sign).
- **Quality as metadata**: per clip the audit's landmark metrics (dominant-hand tracking rate, longest gap, finger / wrist jitter, face visibility, span, swaps, zoom) and the retargeting error are in the index and the IR.
- **Provenance box** (page, bottom left) and IR `provenance`: origin, dataset, clip id, signer, tier, quality, fidelity, repairs ("palm contact: 13 frames of the hidden hand mirrored", "face depth: hand moved 10.3 cm towards the face"), licence, source link; plus the versions behind the sentence.

dataset	licence recorded in provenance
TTRS	TTRS Thai Sign Language dictionary (NECTEC / Ratchasuda) — public dictionary, research use
th-sl.com	rights reserved; research use approved by the project owner, not redistributable
TSL51	Hugging Face dataset licence, research use
YouTube (agent / harvest / phrases)	publisher's rights, research use only (eye-checked)

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5. Motion

5.1 Two v1 defects that only the v2 tests revealed

1. **Thumbs in the browser.** three-vrm renames VRM 0.x thumbs on load (Proximal→Metacarpal, Intermediate→Proximal); v1 wrote its thumb rotations to the names it knew, so the first rotation landed on the second joint and the second was dropped — every handshape with a thumb was wrong **on screen** while all Python metrics (which use the correct rig) looked fine. One mapping table now serves the rig loader, the browser and the `.vrma` export (`rig.TO_VRM1`).
2. **Hands floating in front of the face** — §5.3.

5.2 Palm-to-palm contact (gap [3])

Root cause, found frame by frame on ขอบคุณ: while the palms are pressed together MediaPipe reports **one merged hand twice** (identical landmarks); the cleaner correctly removed the duplicate, so the left hand vanished exactly during the wai and the avatar's hand drifted and crossed. v2 reconstructs the hidden hand as the **mirror image** of the tracked one about the midline between the two pose wrists — **only** when the evidence says the sign is symmetric contact (the pose wrists touch and, in the frames just before AND after, both hands were tracked as mirror images, shape error < 0.35 palm lengths). On those frames the palms are constrained to face each other and the wrists keep ≥ 5 cm apart (two hand thicknesses). 239 clips (1,267 frames) qualified; each says so in its provenance. Result (ขอบคุณ): palm normals **8° apart** (v1: crossed, one usable frame); properly tracked contact signs are untouched (ไหว้ 13° vs signer 16°, ขอโทษ 39° vs 42°).



Figure — v1 vs v2, same moment, front and side view: ขอบคุณ (wai) and ห้องน้ำ (pinch at the nose). The side view shows what the front camera hides.

5.3 Face-contact depth

MediaPipe's pose depth is unreliable; v1 put face-located hands **13–21 cm in front of the face** (ห้องน้ำ's nose pinch 18.7 cm, หน้ากาก 20 cm, กิน 13 cm) — invisible from the front, wrong in 3D. Across the dictionary, runs of frames in which a hand overlaps the face sat 7–22 cm in front (median 14 cm): a systematic bias, not real variation. v2 moves each such run in depth by **one constant** so its closest approach is 3 cm in front of the nose plane, **capped at 15 cm** — a few signs really are made in front of the face (a pane of glass, a book) and must not be pulled onto it; monocular landmarks cannot tell them apart (apparent hand size was tested as a depth cue and did not separate them). The recording's own depth motion inside a run is kept. 5,111 clips have such a run (median shift 14 cm; 2,255 at the cap). The 2D retargeting error barely moves (wrist median 0.067 → 0.070 shoulder widths).

5.4 Transitions and co-articulation (gap [6])

The user accepted "correct and smooth". v2 keeps the stroke of every sign **bit-exact** (test `test_strokes_are_not_changed`: arm and finger rotations of each stroke frame equal the library clip) and changes only transitions: the fingers run on a compressed clock and reach the next handshape at 75 % of the transition (anticipatory handshape, as fluent signers pre-shape the hand). Whole-sentence integrity, 96 Thai sentences generated from words (`t2s/eval/motion_quality.py`):

per 96 sentences (9,069 frames)	v1	v2
hand points inside the torso — stroke / transition frames	49 / 17	51 / 20
hand points inside the head	0 / 0	0 / 0
hands passing through each other — stroke / transition	37 / 4	**24 / 3**
finger flicker (jump > 25° and back)	1	1
body bone > 35° in one frame	17	17
median transition jerk (m/s³)	114.6	116.1
median sentence length	3.66 s	3.70 s

The 17 fast frames are all **inside recorded strokes** that the tempo model plays 1.5× faster (the library limits 30°/frame; ×1.5 = 45°), e.g. tapping the wrist in เวลา — not tracking flips; transitions have none above 35.5°. Torso "penetrations" are mostly finger tips touching the chest in contact signs (my torso box is a crude cuboid).

5.5 Non-manual features (gap [5]) — measured, scoped, cited

`t2s/eval/nmm_data.py` on the 252 TSL51 sentence takes (one signer, gold order) and the 40 recorded conversational phrases (several signers); per clip head yaw / pitch from the face landmarks and brow blendshapes relative to the clip's own neutral.

mark	data	decision
WH question	brows by fifth of the sentence: WH 0.002 · **0.054 · 0.048 · 0.062 · 0.050** vs statements 0.015 · 0.015 · 0.016 · 0.024 · 0.049 (15 vs 72)	brow raise **from the 2nd sign to the end** — not on the first sign, not over the whole sentence
yes/no question	4 recorded yes/no phrases vs 19 statements: head pitch forward at the end p = 0.03, brows up p = 0.09	forward tilt + brow raise on the **last sign**; flagged *weak data*
negation	TSL51 has **no** negative sentence; dictionary negation signs (109 clips, 12 signers) shake the head **2×** more than other signs ($p = 6{\cdot}10^{-17}$), ~5° at ~2.5 Hz, but 61 % only weakly	head-shake on the **negation sign itself**, topped up to ~5° only where the clip's own shake is weaker; no spreading over the predicate (no sentence data)
topic (object first)	165 topic-first vs 72 subject-first sentences: brow on the first sign 0.0136 vs 0.0146, **p = 0.53**	**not applied** — recorded in the IR as "no evidence"

Face track: mouth channels stay **lexical** (each sign's own mouthing); sentence marks touch only brows (max with the lexical value) and the head (composed with the lexical rotation), inside their scope. Every applied mark is listed in `ir.nmm` with frames, scope and evidence, and shown in the page (*"ทำไมเรียงแบบนี้"*).

5.6 Machine reading of the avatar (forward recogniser, leave-source-out)

The forward ThaiSLM recogniser (Thai-only landmark encoder) names the sign from landmarks generated from the avatar's skeleton, with the source clip removed from its bank (it must look like *other* people's version of the sign). Same 1,000 clips, v1 and v2 libraries:

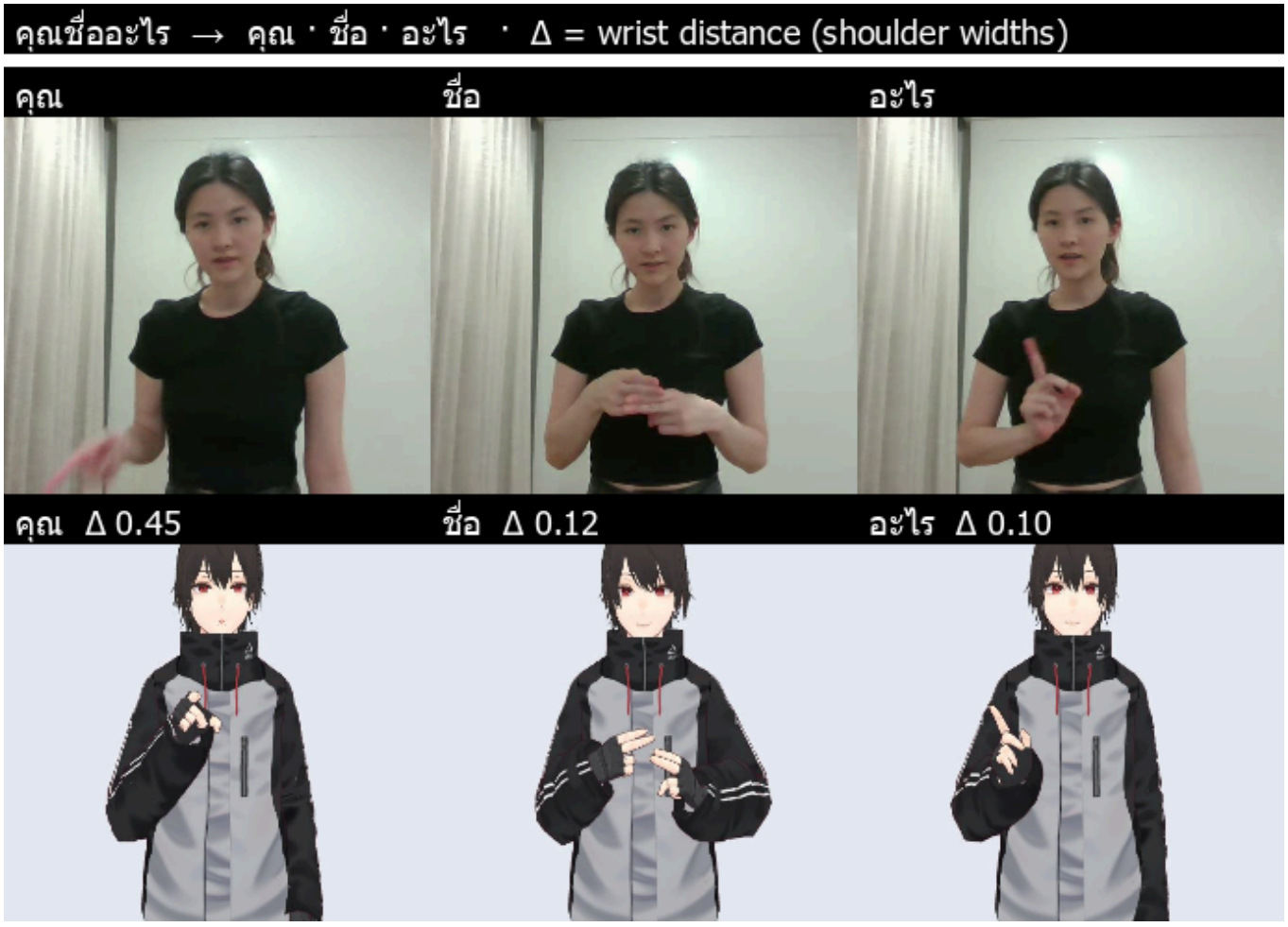
n = 1,000	top-1	top-5	top-10
real signer's landmarks (upper bound)	63.0 %	70.9 %	74.5 %
avatar, v1 library	52.5 %	62.7 %	67.8 %
avatar, v2 library	**53.3 %**	**62.9 %**	**68.2 %**

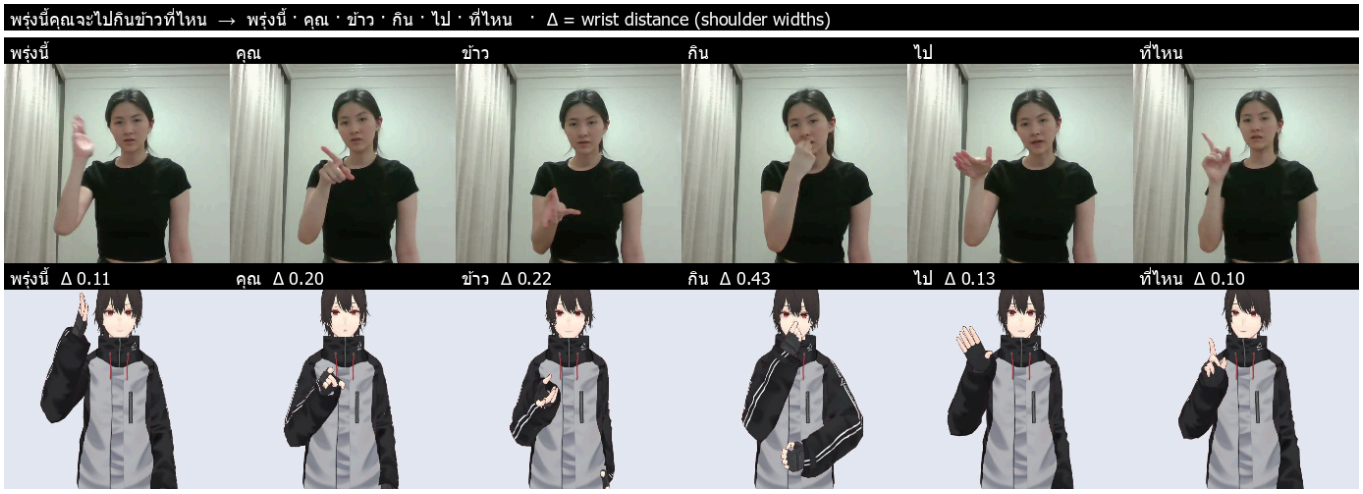
The contact and depth repairs did not cost recognisability (84.6 % of the human top-1, v1 83.3 %). The recogniser reads a front view, so it cannot reward the depth fix — the side views of §5.2 show that.

5.7 Real signer vs avatar — aligned sign by sign (the 5 requested images)

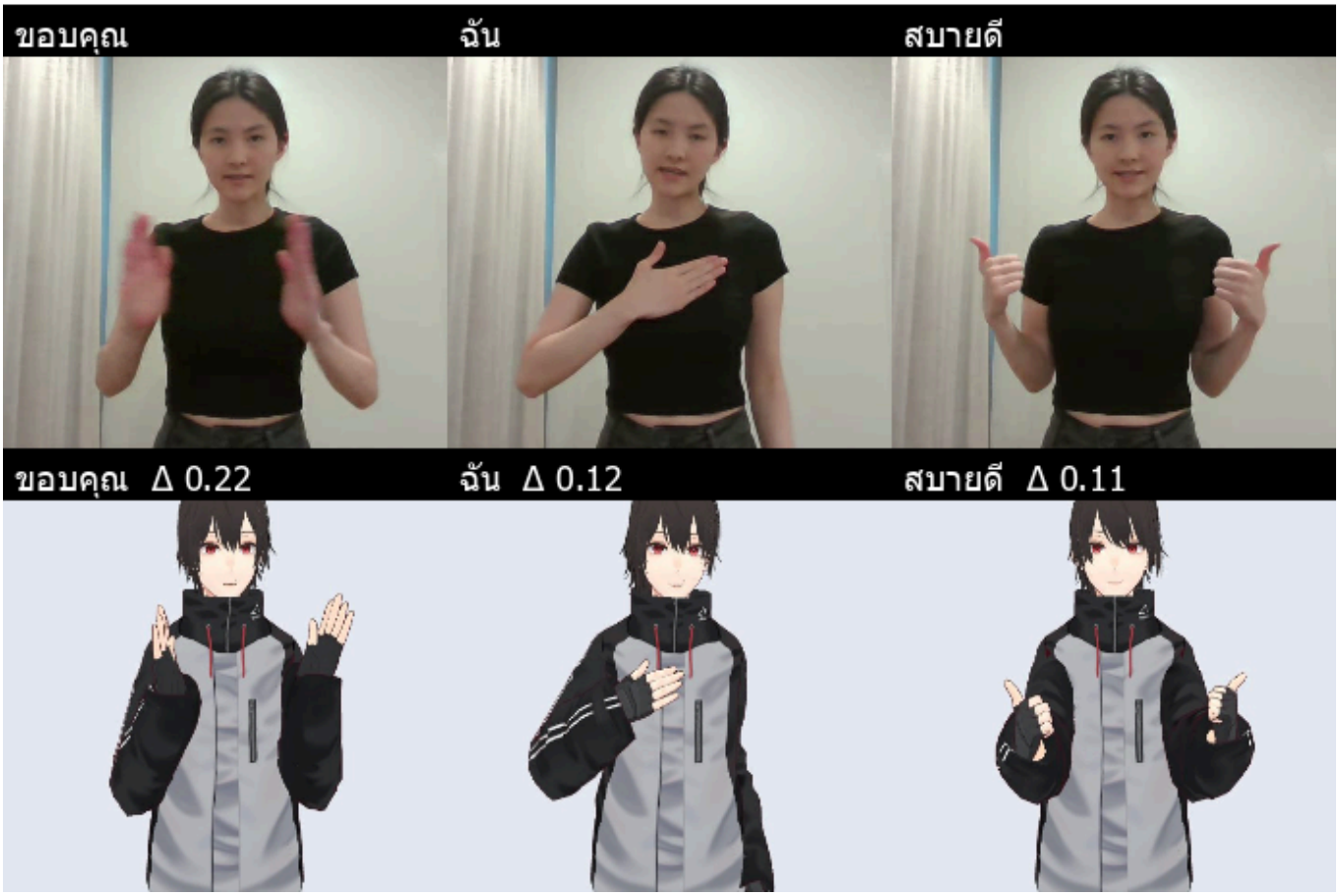
v1's sheets showed evenly spaced frames, so a column could show the person at one sign and the avatar at another. v2 aligns them: the avatar signs the sentence **from words** (the recorded phrase is never shown as the avatar) and the real recording is aligned to it by **dynamic time warping** on what both share (wrists relative to the shoulders, dominant finger tips relative to the wrist). Each column = the middle of one avatar sign (bottom) and the real frame matched to it (top); Δ = wrist distance at that moment (shoulder widths). Four sentences are TSL51 (the same signer recorded the words in our library and the sentence: differences show our system, not a variant choice); the fifth is another signer at conversational speed (the hard case).

#	sentence	glosses (avatar)	mean Δ	avatar / real length
1	คุณชื่ออะไร	คุณ · ชื่อ · อะไร	0.22	2.7 s / 6.2 s
2	พรุ่งนี้คุณจะไปกินข้าวที่ไหน	พรุ่งนี้ · คุณ · ข้าว · กิน · ไป · ที่ไหน	0.20	6.5 s / 13.2 s
3	ขอบคุณ ฉันสบายดี	ขอบคุณ · ฉัน · สบายดี	0.15	4.3 s / 7.2 s
4	ฉันชอบเที่ยวกรุงเทพ	กรุงเทพ · ฉัน · เที่ยว · ชอบ	0.21	5.5 s / 10.2 s
5	คุณกินข้าวหรือยัง (other signer)	คุณ · ข้าว · กิน · ยัง	0.22	4.5 s / 1.8 s



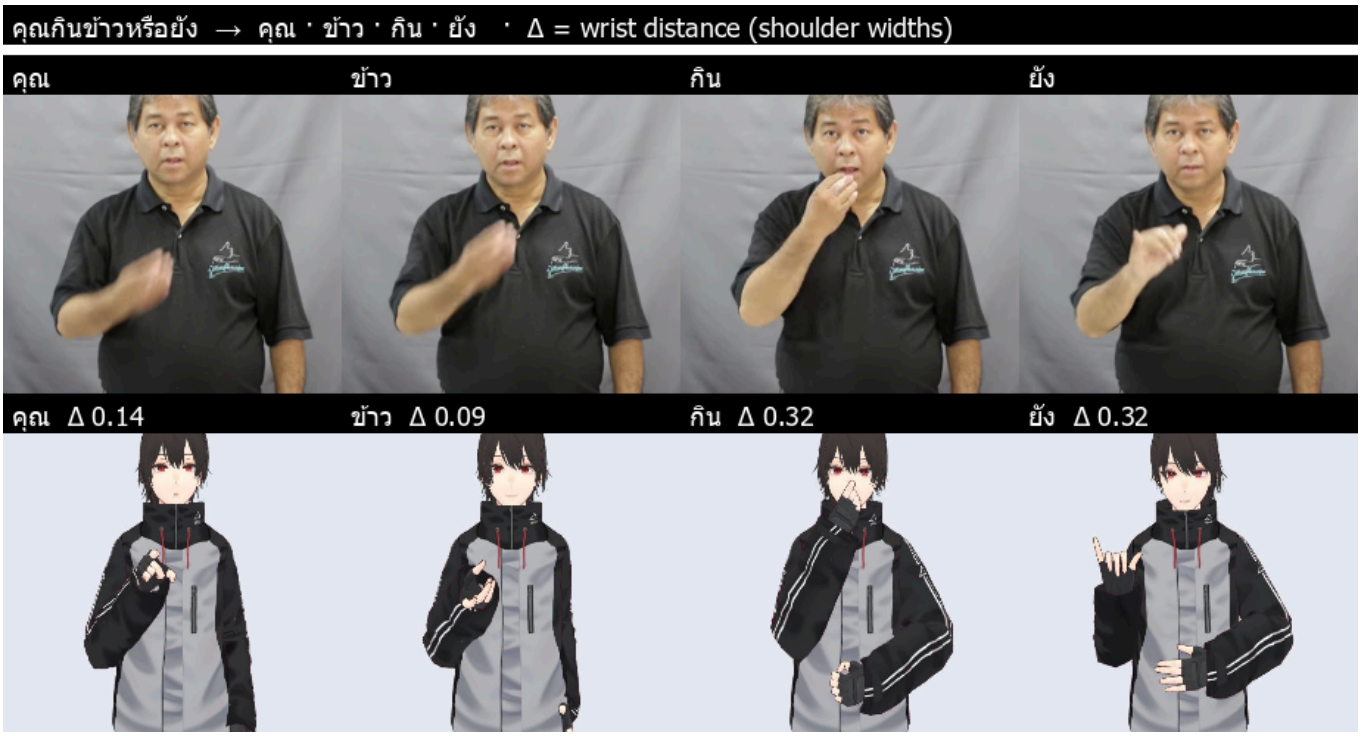


ขอบคุณ ฉันสบายดี → ขอบคุณ · ฉัน · สบายดี · Δ = wrist distance (shoulder widths)



ฉันชอบเที่ยวกรุงเทพ → กรุงเทพ · ฉัน · เที่ยว · ชอบ · Δ = wrist distance (shoulder widths)





What they show: the gloss order of the TSL51 signer and of our rules agree in all four TSL51 sentences; hand shapes and places match where the variant is the same (ฉัน, เที่ยว, ชอบ, สบายดี); the largest Δ are a different place (คุณ: the signer points lower), a different variant (กิน), or timing (the lab recordings are 2× slower than our tempo; the YouTube signer is 2.5× faster).

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6. Characters — every `.vrm` in `vrm/` (gap [7])

v1 handled VRoid exports only. The two new models are **VRM 1.0 exported from Blender**: bones carry rest rotations (v1 refused them) and there is no `upperChest`. v2: the rig loader composes the full node transforms; absent optional bones are **folded** (their rotation moves to the children so the bones that exist keep the library's world rotations); thumbs use the VRM 1.0 names; the body fit uses per-region proportions (face anchors scale with head size, torso anchors with shoulder width sideways — gojo's shoulders are 1.28× the library avatar's). A new file is **checked on arrival** and the result is in `/api/avatars` and the picker tooltip:

character	VRM	absent bones	face channels (direct / via whole-face expression / not shown)	placement error, copy → fit (cm)	licence (from the file)
Default (AvatarSample_C)	0.x	—	12 / 0 / 0	library avatar (identity)	allow
Model 1	0.x	—	12 / 0 / 0	3.2 → 0.8	allow
Model 2	0.x	—	12 / 0 / 0	4.3 → 1.7	**commercial: disallow**
Model 3 (gojo)	**1.0**	upperChest	0 / 10 / 2	9.0 → 4.2	personal non-profit
Model 4 (yuta)	**1.0**	upperChest	0 / 10 / 2	9.0 → 3.8	personal non-profit

Placement error = how far the hand lands from where the library avatar puts it relative to the nearest body part, scaled to the body (40 everyday sentences); the rest comes from genuinely different proportions (wide shoulders) where "the same place" is not uniquely defined. Hand orientation is identical (< 0.5°) for all. Tested in pytest for every file in `vrm/`, so any model added later is covered without code changes.

Picker thumbnails: VRoid files embed one; the two Blender VRM 1.0 files do not. A character without one gets a head-and-shoulders image rendered once by the browser in the background and stored by the API in the avatar cache (`POST /api/avatars/{id}/thumbnail`, never overwriting an existing thumbnail), so every later visitor sees it immediately.

Default · ห้องน้ำ



Default · อยู่



Default · ที่ไหน



Model 1 · ห้องน้ำ



Model 1 · อยู่



Model 1 · ที่ไหน



Model 2 · ห้องน้ำ



Model 2 · อยู่



Model 2 · ที่ไหน



Model 3 · ห้องน้ำ



Model 3 · อยู่



Model 3 · ที่ไหน



Model 4 · ห้องน้ำ



Model 4 · อยู่



Model 4 · ที่ไหน





Figure — all five characters signing ห้องน้ำอยู่ที่ไหน at the same moments.

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7. Speech input (gap [2])

7.1 Test audio — same for every configuration, development and test kept apart

set	what	role
dev	v1's 96 everyday sentences, VachanaTTS female voice	tuning
test-A	the 73 held-out Thai sentences, VachanaTTS **male** voice	test
test-B	20 long utterances (3–5 test-A clips with natural pauses, median 8 s)	test (chunking)
FLEURS	100 utterances of Google FLEURS Thai — **real human speech**, long read sentences	test (real speech)

7.2 Model × device (local, $0; same audio)

CER / latency	small · CPU	small · GPU	medium · CPU	medium · GPU
dev (TTS female)	16 % (v1)	20.5 %	—	15.0 %
test-A (held-out, male)	2.5 %	2.5 %	**0.9 %**	1.4 %
test-B (long)	1.2 %	1.0 %	1.2 %	1.2 %
FLEURS (real speech)	—	**6.1 %**	—	**3.8 %**
latency, short utterance (p50)	2.4 s	0.36 s	**6.9 s**	**0.71 s**
latency, 8 s utterance (p50)	3.3 s	1.0 s	9.7 s	1.8 s
VAD pause-chunking (FLEURS)	—	6.6 % (worse)	—	4.9 % (worse)

Decisions: **GPU → medium, CPU → small** (automatic, `SPEECH_DEVICE=auto`); bigger is not always better — on CPU medium costs 2.9× the latency for ~1.5 points. **Chunking at pauses makes real speech worse** (the model loses the sentence context) — it is kept only for the live partial text; the final text re-decodes the whole utterance when it is ≤ 12 s.

Where measured: the GPU columns ran on the local RTX 2050 **outside Docker** (CTranslate2 4.5 with CUDA 12 / cuDNN 9). The deployed stack on this laptop runs the CPU path. The GPU container variant (`service/speech/compose.gpu.yml`, build arg `WITH_CUDA=1`) is provided but **was not built or run here** (it downloads ~2 GB of CUDA libraries and needs Docker GPU support) — treat it as untested until it is run once.

7.3 Why v1 measured 16 %: invented greetings, and the reliability check

Almost all of dev's errors are **0.2–0.5 s clips** for which the Thai fine-tuned Whisper invents "สวัสดีครับ สวัสดีครับ" or "ขอบคุณที่ติดตาม" (subtitle phrases from its training videos). Padding with silence did not help (0/23 fixed). v2 does not sign such text: a transcript is flagged when it is physically impossible (> 25 characters per second of speech), repeats itself inside a short utterance, is a stock subtitle phrase on < 1.2 s of speech, empty, or has very low recogniser confidence. Offline on GPU runs it caught 10 of 23 catastrophic dev transcripts with 1 false alarm, and **0 false alarms** on test-A / test-B. The page then says *"ระบบไม่แน่ใจ…"* with the reasons and offers *พูดใหม่* / *ใช้ข้อความนี้*. Latency guard: such hallucinations took 30–45 s on the CPU (all temperature fallbacks) and blocked the queue — generation is now capped by the audio length.

7.4 Context-aware correction — evaluated, shipped as suggestions

`t2s/text/asr_correct.py` changes a span only with all four kinds of evidence: it **sounds** the same (Thai soundex) **and** reads almost the same (≤ 30 % characters), the replacement is a **signable** word, the Thai National Corpus bigrams make the sentence ≥ 50× more likely, and (when available) the recogniser was unsure. Thresholds set on dev only.

set	CER before → after	fixed	harmed
dev	20.5 → 20.3 %	2	0
test-A (held-out)	2.47 → 2.47 %	0	0
test-B (long)	1.0 → 1.0 %	0	0
FLEURS (real speech)	**6.10 → 6.34 %**	7	**24**

On real speech it damaged transliterated names more often than it fixed misspellings, so automatic correction would break the "only with sufficient evidence" rule. It is shipped as **suggestions** (*"แนะนำแก้: X → Y … [ใช้]"*), applied only by the person; `T2S_ASR_CORRECT=auto` exists for experiments. The language model is a 5.6 MB array file (58 MB in memory instead of 0.6 GB as Python dicts).

7.5 Live recognition through the stack

Measured through nginx → API → speech container (CPU, Whisper-th small, as deployed without a GPU):

set	CER	exact	wait after speaking (p50 / p95)	wrong transcripts (CER > 0.3) caught by the reliability check	false alarms
dev (v1's 96)	16.2 % (v1: 16 %)	64 / 96	1.97 / 2.17 s	**11 of 23**	1
test-A (held-out)	**2.05 %**	62 / 73	2.09 / 2.48 s	0 of 0	0
test-B, streamed in real time	**0.94 %**	13 / 20	3.14 / 3.81 s (8 s utterances)	0 of 0	0

If the person re-types or repeats the flagged dev transcripts, dev CER falls from 16.2 % to 7.4 %. While speaking, the page shows committed text (closed by a pause) and grey tentative text, updated every second; the final text is a full re-decode for ≤ 12 s of speech (chunking at pauses was measured worse), so the wait after speaking equals a one-shot request — the gain is that the person sees and can start checking the text while still speaking (`SPEECH_FULL_FINAL_S` trades accuracy for a shorter final wait on long utterances).

Two defects of my own were found by this system test and fixed before the numbers above: a too-tight token cap truncated sentences, and the decoding lost v1's *no timestamps* setting (Whisper then started a second segment and repeated the sentence) — held-out CER through the stack was 7.3 % until both were fixed; they would not have shown in the offline runs.

Separation of errors (prompt §E): the IR keeps `input.asr = {raw, model, reliable, flags, suggestions, applied, edited}` next to the text actually translated; the translation confidence never uses the recogniser's confidence.

End-to-end UI test (headless Chromium with a WAV file as the microphone, `--use-fake-device-for-media-stream`): partial requests every second, the final text filled into the box, the reliability box and suggestions rendered, signing started — all working. Found: the final request waits behind a partial that is still being decoded (one lock per speech container), adding up to ~3 s on a CPU; a dedicated partial worker or a GPU removes it (next step).

After user testing (2026-10-11): the interim box was unlabelled and once showed an invented "สวัสดีครับ สวัสดีครับ" — the quick interim decode retried without the voice detector on a silent tail. Interim text now runs only where speech is detected and passes the same reliability check as the final text (tested: no invented greeting on the short dev clips; real speech still streams in second by second); the box is labelled *ข้อความชั่วคราวระหว่างพูด (ยังไม่ใช่ผลสุดท้าย)*, and a real-time progress bar (green → indigo) shows while the final text is recognised: the recogniser cannot report its own progress, so the bar runs on the time this recording is expected to take, from the latencies measured above (CPU ≈ 1.8 s + 0.16 s per second of audio — 1.97 s measured for 1 s, 3.1 s for 8 s; GPU ≈ 0.45 s + 0.17 s/s), eases to 92 % at that time, creeps if it runs longer and completes when the text arrives, with the percentage and "elapsed / expected" seconds.

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8. System performance

Deployed stack on the laptop CPU (docker, 2 API workers), sentences the API had not cached:

path	p50	p95	note
Thai text → IR + motion, short sentence (through nginx)	**53 ms** (v1: 59 ms)	98 ms	language stage 1.0 ms, motion 33 ms; 20 kB gzip per sentence (v1: 28 kB)
Thai text → IR + motion, long sentence (8–13 s of signing)	100 ms	151 ms	motion 64 ms
English text → NLLB → IR + motion	**332 ms** (v1: 497 ms)	396 ms	MT 278 ms
4 concurrent clients	89 ms	305 ms	23 requests / s
8 concurrent clients	201 ms	678 ms	23 requests / s (CPU-bound, 2 workers)
Thai speech → text (CPU, small)	2.0–2.1 s	2.5 s	§7.5
Thai speech → text (GPU, medium)	0.7 s	1.2 s	local RTX 2050, §7.2
first load	app 203 kB gzip (v1: 195 kB) · avatar 3.5 MB		other characters: optimised copy on first use
memory	API 0.99 GB (2 workers) · speech 1.34 GB · web 11 MB		the ASR language model was cut from 0.6 GB to 58 MB per worker
browser, JavaScript per frame	**0.10 ms**		v1: 0.67 ms (timeline sample + bone / morph writes; GPU excluded)

Latency did not regress; English measured faster than in v1 with the same model (cause not isolated). Load first: the speech container needs ~10 s to load the models; the API ~3 s.

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9. Data collection (gap [4]) and cost

Coverage. Targets came from the development sentences and corpus frequency only (not the held-out set): ทำ เสื้อ ราคา ลดราคา จริง ทุก โมง เรื่อง ทาง เมือง ตาย แดง โรค ช้าง รอบ แทน เหลือ แก้ ชั้น คืน…. Searches: v1's harvest queries for 29 words (4 exact lessons, 3 for words already held), then 4 phrasings × 25 results for the 8 core words (800 results, **1** exact lesson: เสื้อ, signed by a school student — rejected under v1's *no child learners* rule, listed for expert review), and a channel of short daily-word clips (47 clips, all already in the dictionaries). TTRS holds เรื่อง only as a school variant and "ทุก" only in "ทุก ๆ วัน". th-sl.com is not harvested by the agent (robots: no AI agents). **Conclusion: these words need a recording session.**

Cost. Cloud **$0**: no Vertex job, no GCS object, no VM — verified after the work: bucket `temp-gpu-bucket-job` empty, 0 custom jobs in us-central1 / asia-southeast1 / europe-west4, 0 Compute instances. GPU experiments on the local RTX 2050. Downloads: FLEURS Thai dev (247 MB, free), TNC bigrams via pythainlp (42 MB). Agent usage is billed by Anthropic and is not visible to me.

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10. Limits and next steps (honest)

1. **Comprehension by Deaf viewers is still unmeasured** — every number here is a proxy. The Grammar Format switch, the provenance box and the variant selector are built for that review.
2. **Grammar generalisation** is unknown beyond TSL51; the held-out set tests consistency, not correctness.
3. **Recording request (≈ 10 words):** ทำ, เสื้อ, ราคา, ลดราคา, จริง, ทุก, โมง, เรื่อง (main form), ตา (eye), กาน้ำ — with one adult Deaf signer, 2–3 takes each; `python main.py harvest`-style QC then `python main.py add-clips` (no full rebuild).
4. **Negation in sentences**: no recorded negative sentence exists; the head-shake is lexical only.
5. **Face depth prior** is a correction of a measured bias, capped; signs truly made in front of the face are moved up to 15 cm.
6. **ASR** on real microphones / accents is unmeasured (FLEURS is read speech); medium needs a GPU for interactive use.
7. Names that are also dictionary words (ส้ม) are signed, not spelled.
8. Licensing: th-sl.com, YouTube and two characters (personal non-profit / no commercial use) — research use only.

11. Reproduce

```
cd reverse_text2sign
python main.py library && python main.py lexicon && python main.py test          # 44 tests
python main.py eval2 language | motion | nmm | compare | asr                       # §3–§7 (asr / compare need the stack)
cd .. && docker compose up --build                                                 # → http://localhost:8081
```

v1 for comparison: `git worktree add ../v1 dfb7496` and run the same evaluators against it (this report did exactly that). File map: `README.md`. Derived data in `data/` (git-ignored).